I. Applications, resources, and decision guideline

A. Downstream applications

(1) Intelligent transportation system. The accuracy of spatiotemporal data is crucial for Intelligent Transportation Systems (ITS) functions like traffic prediction, network optimization, and incident early warnings [1]. Tensor decomposition-based methods are highly effective at modeling the spatiotemporal characteristics of traffic data and addressing data loss issues caused by sensor malfunctions or communication interruptions [2]. For example, Tan *et al.* [3] proposed a dynamic tensor imputation algorithm leveraging multi-modal low-rank properties and pattern similarity modeling, significantly enhancing traffic prediction accuracy under missing data conditions. Tang *et al.* [4] introduced a tensor decomposition approach for signalized intersections, modeling traffic flow distribution similarity using sparsely sampled vehicle trajectory data to provide a novel modeling perspective for traffic state awareness in vehicle-infrastructure cooperative environments. Furthermore, Dong *et al.* [5] presented a Laplacian-integrated method combining graph convolutional networks with tensor imputation for traffic prediction with missing data in intercity highway networks. This approach utilizes graph-based similarity modeling to enhance the accuracy of traffic state prediction for large-scale highway networks, even in the presence of partial input data loss and sparse detector distribution, thereby providing an effective and robust solution.

(2) Environment monitoring. Accurate environmental monitoring and trajectory data are vital for functions such as pollutant tracking, disaster early warning, and ecological preservation. This data, including air quality indices, meteorological elements, and maritime trajectories, forms the basis for environmental assessments [6]. However, environmental data often has gaps due to sensor failures, extreme weather, or human error, while trajectory data can be discontinuous from sparse sampling or device drift. To address these challenges, researchers have proposed tensor decomposition-based methods for imputing missing environmental data. The imputed data are then applied to various downstream environmental monitoring tasks, significantly improving the accuracy and reliability of environmental state assessments. For example, Xu *et al.* [7] utilized tensor decomposition and sequence-based correlation modeling to extract latent information from diverse heterogeneous data sources. The imputed data subsequently served as input for a hybrid prediction model for carbon emissions, leading to more accurate forecasts. Qi *et al.* [8] introduced a three-dimensional tensor method specifically for port ship carbon emissions. This approach leverages sub-tensor correlation modeling to decompose the tensor into multiple sub-vectors, enabling multi-dimensional feature extraction and missing data imputation for ship carbon emissions. This provides crucial data support and a decision-making basis for developing energy-saving and emission-reduction measures for ships, thereby promoting green and sustainable development in the shipping industry.

(3) Climate science. In climate science, the integrity and accuracy of spatiotemporal data are essential for climate pattern discovery, extreme weather event prediction, carbon cycle simulation, and climate change impact assessment [9]. However, climate data frequently suffer from missing values due to factors like sparse observation stations, equipment malfunctions, or data transmission interruptions [10]. Tensor decomposition-based techniques for missing data imputation are highly effective in processing high-dimensional, heterogeneous spatiotemporal climate data. These techniques can recover the intrinsic structure and dynamics of such datasets, thereby providing more complete and reliable inputs for climate models. This is crucial for improving the accuracy of climate predictions, gaining deeper insights into the climate system, and formulating strategies to address climate change. For example, Xu *et al.* [11] proposed a supervised tensor decomposition-based multi-task learning method for climate data imputation and prediction. This method encodes climate data as a third-order tensor through feature-based similarity modeling, thereby extracting interpretable spatial and temporal latent factors. Its effectiveness has been validated using global-scale climate datasets. To address the pervasive heterogeneity in climate data, Li *et al.* [12] introduced an improved tensor feature analysis method. This approach involves dividing and reconstructing heterogeneous data into sub-tensors and employing tensor decomposition for feature analysis, effectively mitigating traditional biases. Additionally, it can reveal latitudinal changes in temperature gradients and anomalies in air temperature, significantly enhancing data approximation accuracy in complex regions and during climate anomaly events.

(4) Other applications. Beyond intelligent transportation, environmental monitoring, and climate science, tensor decomposition-based spatiotemporal data imputation has vast applications across diverse fields [13]. In the domain of shared mobility demand prediction, for instance, Yan *et al.* [14] proposed HT-STNet, which combines a tensor decomposition model with an LSTM-based sequence similarity modeling approach. This effectively addresses the challenges of modeling high-dimensional, sparse spatiotemporal data. Its capabilities in dynamic sparsity awareness, multi-scale dimensionality reduction, and lightweight modeling offer an efficient and robust solution for tasks like shared mobility demand forecasting. In the context of energy systems, Madbhavi *et al.* [15] introduced a novel state imputation framework based on block tensor imputation. The primary objective is to capture the spatiotemporal characteristics of power system states through tensor modeling and optimize the imputation process by incorporating physical constraints. Moreover, in the field of medical and health monitoring, Cong *et al.* [16] employed tensor decomposition to impute electroencephalogram (EEG) signals and extract signal features. This approach assists in uncovering physiologically significant patterns from high-dimensional, sparse EEG signals, thereby providing a powerful tool for medical diagnosis, brain-computer interfaces, and cognitive science.

B. Code and datasets

Tensor decomposition-based methods have emerged as a cornerstone for research and applications in spatiotemporal

TABLE I: Links to the publicly available dataset

Data Classification	Dataset Name	Data Type	Sampling Rate	Data Dimension	Dataset Links
Time-dynamic, spatial-static (Node*Day*Time)	PeMS dataset	Traffic flow	5 minutes	358*91*288 (PEMS03) 307*59*288 (PEMS04) 170*62*288 (PEMS08)	http://pems.dot.ca.gov/
	GTL dataset	Traffic flow/speed/ occupancy	5 minutes	26*31*288	https://gtl.inrialpes.fr/
	Hangzhou dataset	Metro inflow/outflow	10 minutes	81*25*144	https://tianchi.aliyun.com/.../information
	Hangzhou metro passenger flow dataset	Metro inflow/outflow	10 minutes	81*25*144	https://zenodo.org/record/3145404
	Portland	Traffic flow/speed/ occupancy	5 minutes	156*31*288	https://new.portal.its.pdx.edu/?/home
	Seattle	Traffic speed	5 minutes	323*365*288	https://github.com/zhiyongc/Seattle-Loop-Data
	Shenzhen Metro Data	Metro inflow/outflow	10 minutes	172*31*144	https://github.com/yongmeizhao/smartcard-data/blob/main/data.rar
	METR-LA	Traffic speed	5 minutes	207*121*288	https://paperswithcode.com/dataset/metr-la
	GuangZhou traffic speed dataset	Traffic speed	10 minutes	214*61*144	http://www.openits.cn/openData2/792.jhtml
	Birmingham parking dataset	Parking occupancy	30 minutes	30*77*18	https://archive.ics.uci.edu/parking+birmingham
	Bond-Fund holding EPS Fundamentals	Bond excess returns & chars	1 month	8759*206*29	https://sethpruitt.net/2022/03/29/reconciling-trace-bond-returns/
	ShenZhen traffic speed dataset	Traffic speed	15 minutes	156*31*96	https://github.com/lehaifeng/T-GCN/tree/master/data
	Xi'an traffic speed data	Traffic speed	10 minutes	500*30*144	sts.didiglobal.com
	PM10 (London)	Air Quality	Hourly	72*90*24	https://zenodo.org/records/4531304
	Beijing Transportation Smart Card data	Metro inflow/outflow	10 minutes	276*31*144	https://github.com/yongmeizhao/smartcard-data/blob/main/data.rar
Time-dynamic, spatial-dynamic (Node*Time*Attribute)	Xi'an trajectory (Didi)	Vehicle Trajectory	1 minute	500*1440*3	https://sts.didiglobal.com/
	Air track	Flight trajectory	1 minute	1000*1440*5	https://zh.flightaware.com/
	US Coastal AIS	Vessel Trajectory	1 minute	2000*1440*4	https://marinecadastre.gov/accessais/

data imputation. With increasing theoretical advancements and practical demands, a wealth of high-quality public datasets and open-source code resources has become available. These resources provide robust support for algorithm validation, performance comparison, and cross-domain transfer. To systematically organize the currently available data resources and implementation tools, this survey compiles representative public datasets alongside their code implementations, aiming to offer a unified reference framework for future research. Presently, datasets and codes suitable for tensor decomposition-based spatiotemporal data imputation tasks encompass various application scenarios, including traffic flow, subway passenger flow, parking status, financial data, and trajectory information. Based on whether their spatial structure changes over time, these datasets can be categorized into two main types: time-dynamic, spatial-static, and time-dynamic, spatial-dynamic, as illustrated in Tab. I and II.

C. Comparative performance synthesis

This paper provides a systematic classification of representative works in tensor imputation through the lens of spatiotemporal effect modeling. It summarizes the advantages and limitations of various methods, alongside their applicability across different missing patterns (RM/BM/TM/MM), offering a strategic reference for model selection in diverse data scenarios. As detailed in Table III, we establish a comparative framework based on the triad of "spatiotemporal effect—methodological category—missing pattern". Methods that model spatiotemporal correlation (SC) often rely on static regularization terms, such as low-rank decomposition, tensor nuclear norm (TNN), graph Laplacian regularization, or fixed convolutional kernels. By leveraging local continuity and topological constraints to assist the optimization of the objective function, these methods are generally more robust in scenarios involving RM and BM. However, when data are characterized by TM patterns that cause discontinuities along the time axis, the effective range of information propagation for these static regularizers becomes significantly restricted, which in turn can lead to performance degradation. In contrast, methods that characterize spatiotemporal heterogeneity (SH) employ dynamic and adaptive distribution or representation learning mechanisms (e.g., fast adaptation in meta-learning, generative priors) to mitigate the systematic bias introduced by the global homogeneity assumption. Consequently, they are more likely to maintain robustness under TM and certain MM conditions. However, their efficacy often depends on factors such as the scale of the training data, the appropriateness of the task formulation, and the quality of multi-source data alignment. Approaches based on geographical similarity (GS) typically leverage external semantic or functional contexts (e.g., POI, land use, road hierarchy) to establish cross-regional

TABLE II: Links to publicly available code

Type	Method	Code Links	Platform
Tensor decomposition-based methods	LRTC-3DST [17]	https://github.com/HaoShu2000/LRTC-3DST	MATLAB
	Meta-TD [18]	https://doi.org/10.6084/m9.figshare.25997134	Python
	BGCP, BTTF, BATF, BTMF, LRTC-TNN [19], [20]	https://transdim.github.io/	Python
	GT-TDI [21]	https://github.com/KP-Zhang/GT-TDI	Python
	CP-ALS-QR [22]	https://github.com/rlminste/CP-ALS-QR	MATLAB
	Hg-TD [23]	https://figshare.com/s/17b6beeb...	Python
Tensor nuclear norm-based methods	MVLR [24]	https://github.com/heyang49/MVLR	Python
	CJSR [25]	https://github.com/xbchen82/CJSR	MATLAB
	LRTC-TMCP [26]	https://github.com/peterchen96/LRTC-TMCP	Python
	ST-LRTC [27]	https://doi.org/10.6084/m9.figshare.20289078	Python
	STH-LRTC [28]	https://github.com/mcgill-smart-transport/traffic_state_estimation	MATLAB
	tensor4kriging [29]	https://github.com/tongnie/tensor4kriging	Python
	LRTC_DeepPnP [30]	https://github.com/peterchen96/LRTC_DeepPnP	Python
	RTC-GTNLN [31]	https://github.com/HaoShu2000/RTC-GTNLN	MATLAB
	STTC-L0-NN [32]	https://github.com/TCCofWANG/STTC-L0...	Python
	ImputeFormer [33]	https://github.com/tongnie/ImputeFormer	Python

associations. This enhancement improves model performance in MM scenarios where observations are extremely sparse or physical topological constraints become unreliable. Nevertheless, the robustness of these methods is more susceptible to issues such as insufficient coverage and noise in the external data.

D. Method selection decision guidelines

To transform the theoretical classification system into an actionable implementation guide, we have integrated the modeling mechanisms and their respective limitations into a priority-based decision framework (Fig. 1). This framework is designed as a "bottleneck-driven" heuristic logic, aimed at identifying an appropriate modeling path based on the primary source of imputation difficulty.

(1) Path selection based on modeling bottlenecks

Users should first evaluate data characteristics to identify the dominant modeling bottleneck:

• **Path A: spatiotemporal heterogeneity (SH) modeling.** This path should be prioritized when the data exhibit strong non-stationarity and local variation, significantly violating global homogeneity assumptions. While generative and multi-view methods are powerful for pattern recovery under extreme sparsity, they require cautious management regarding training instability and the risk of violating physical consistency.

• **Path B: geographical similarity (GS) modeling.** This is the preferred primary path when the intrinsic structural information of the tensor is insufficient (e.g., in scenarios with a high proportion of mixed missing (MM)), but rich and informative external spatial priors (such as POI or road levels) are available. The efficacy of this path is closely tied to the representation quality of external data and its semantic alignment precision with the target variables.

• **Path C: spatiotemporal correlation (SC) modeling.** When the tensor possesses significant intrinsic dependencies—such as mathematical smoothness, periodicity, or clear network topology—this path offers the most efficient and interpretable solution. However, static SC regularization terms may experience substantial performance degradation in long-term missing (TM) scenarios.

(2) Implementation strategy and hybrid extensions

The framework emphasizes that these paths are not mutually exclusive. For complex spatiotemporal datasets exhibiting both heterogeneity and structured dependencies, a "priority-first" hybrid strategy is recommended. In such cases, the core bottleneck should be addressed via the corresponding primary path first (e.g., utilizing SH to capture non-stationarity), followed by the introduction of secondary priors (e.g., GS-based contextual constraints) to further refine the solution space and enhance cross-domain generalization. Consequently, the imputation task shifts from pure algorithmic tuning toward "knowledge engineering", where structured prior knowledge is systematically embedded within the tensor decomposition process.

REFERENCES

[1] I. Laña, I. Olabarrieta, M. Vélez, and J. Del Ser, "On the imputation of missing data for road traffic forecasting: New insights and novel techniques," *Transp. Res. Part C Emerg. Technol.*, vol. 90, pp. 18–33, May 2018, doi: 10.1016/j.trc.2018.02.021.

[2] Y. Berhanu, D. Schröder, B. T. Wodajo, and E. Alemayehu, "Machine learning for predictions of road traffic accidents and spatial network analysis for safe routing on accident and congestion-prone road networks," *Results Eng.*, vol. 23, p. 102737, Sep. 2024, doi: 10.1016/j.rineng.2024.102737.

[3] H. Tan, Y. Wu, B. Shen, P. J. Jin, and B. Ran, "Short-Term Traffic Prediction Based on Dynamic Tensor Completion," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 8, pp. 2123–2133, Aug. 2016, doi: 10.1109/TITS.2015.2513411.

[4] K. Tang, C. Tan, Y. Cao, J. Yao, and J. Sun, "A tensor decomposition method for cycle-based traffic volume estimation using sampled vehicle trajectories," *Transp. Res. Part C Emerg. Technol.*, vol. 118, p. 102739, Sep. 2020, doi: 10.1016/j.trc.2020.102739.

[5] H. Dong, F. Ding, H. Tan, and H. Zhang, "Laplacian integration of graph convolutional network with tensor completion for traffic prediction with missing data in inter-city highway network," *Phys. Stat. Mech. Its Appl.*, vol. 586, no. 1, p. 126474, Jan. 2022, doi: 10.1016/j.physa.2021.126474.

[6] F. A. Y. N. Schröder, D. H. P. Turban, A. J. Musser, N. D. M. Hine, and A. W. Chin, "Tensor network simulation of multi-environmental open quantum dynamics via machine learning and entanglement renormalisation," *Nat. Commun.*, vol. 10, no. 1, p. 1062, Mar. 2019, doi: 10.1038/s41467-019-09039-7.

[7] X. Xu, X. Cao, and L. Yu, "Carbon emissions forecasting based on tensor decomposition with multi-source data fusion," *Inf. Sci.*, vol. 681, no. 1, p. 121235, Oct. 2024, doi: 10.1016/j.ins.2024.121235.

[8] Y. Qi, J. Yang, and K. S. Qin, "Spatial-temporal analysis of carbon emissions from ships in ports based on AIS data," *Ocean Eng.*, vol. 308, no. 1, p. 118394, Sep. 2024, doi: 10.1016/j.oceaneng.2024.118394.

[9] J. Sillmann et al., "Understanding, modeling and predicting weather and climate extremes: Challenges and opportunities," *Weather Clim. Extrem.*, vol. 18, pp. 65–74, Dec. 2017, doi: 10.1016/j.wace.2017.10.003.

TABLE III: Comparative synthesis of representative tensor-based spatiotemporal imputation methods

Effect	Model	Method	External data required	RM	BM	TM	MM	Strengths	Limitations
SC	Feature-based modeling	CRS [34], MRTC [35], MTC [36], TGMC [37], CP-wALS [38], BNPTD [38], LRTC-SCAD [30], TNN-HTV [40], CTMTR [41], CJSR [42], LTRR-NLS [43], POD-DEIM [44], GLOSS [45], Reg. LRTC [46]	No	○	✓	○	○	Simple, interpretable; strong for periodic data.	Fails under strong non-stationarity; sensitive to noise.
	Physics-based modeling	NT-DPTC [47], HRST-LR [48], GCLSNL [49], FDPTC [50], ST-LRST [51], LRTC-3DST [52], NTF [8], RTTC [53], LCR [54], RTLRR [55], GLOSS [56]	Yes	✓	✓	○	○	High interpretability; robust spatial smoothness.	Requires rigid, accurate physical/structural priors.
	Sequential-based modeling	STLSTM [57], RegTensor [58], LTP [59], NTC [60], LSTM-GL-ReMF [61]	No	○	○	✓	✓	Captures nonlinear temporal evolution (e.g., LSTMs).	High training cost; potential for instability.
	Graph-based modeling	GT-SVD [62], GCRINT [63]	Yes	✓	✓	○	○	Superior for explicit network topologies (e.g., roads).	Sensitivity to graph quality and misspecification.
SH	Meta-learning based modeling	Meta-TD [18]	Yes	✓	✓	✓	○	Adaptive spatial weighting; good for non-stationarity.	Highly sensitive to task/meta-knowledge quality.
	Multi-view based modeling	HAN [64], MATCN [65], SMV-NMF [66]	Yes	✓	✓	✓	○	Fuses diverse perspectives to reduce single-view bias.	Depends on cross-view alignment and consistency.
	Generative-based modeling	ST-GAIN [67], DATC [68], CA-GAN [69]	Optional	○	○	✓	✓	Synthesizes patterns under extreme sparsity (e.g., GAN).	May generate implausible values without strong constraints.
GS	Context-based modeling	CARTE [70], [71], NR-cNTF [72], STDI [73]	Yes	○	✓	✓	✓	Enables knowledge transfer via external semantics.	Reliance on high-quality external multi-source data.
	Encoding-based modeling	GT-TDI [21], TRTF [74]	Yes	○	✓	✓	✓	Captures functional consistency via embeddings.	Performance limited by encoder representation quality.
	Dynamic graph-based modeling	ST-LRTC [75]	Yes	✓	✓	✓	✓	Adapts to evolving topological similarities.	High computational overhead for graph updates.

Note: ✓ = Recommended; ○ = Conditional; × = Not Recommended.

[10] C. Li, X. Ren, and G. Zhao, "Machine-Learning-Based Imputation Method for Filling Missing Values in Ground Meteorological Observation Data," *Algorithms*, vol. 16, no. 9, Art. no. 9, Sep. 2023, doi: 10.3390/a16090422.

[11] J. Xu, J. Zhou, P.-N. Tan, X. Liu, and L. Luo, "Spatio-Temporal Multi-Task Learning via Tensor Decomposition," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 6, pp. 2764–2775, Jun. 2021, doi: 10.1109/TKDE.2019.2956713.

[12] D. Li, Z. Yu, F. Wu, W. Luo, Y. Hu, and L. Yuan, "The Tensor-based Feature Analysis of Spatiotemporal Field Data With Heterogeneity," *Earth Space Sci.*, vol. 7, no. 2, p. e2019EA001037, 2020, doi: 10.1029/2019EA001037.

[13] L. Cai, H. Wang, C. Sha, F. Jiang, Y. Zhang, and W. Zhou, "The Mining of Urban Hotspots Based on Multi-Source Location Data Fusion," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 2, pp. 2061–2077, Feb. 2023, doi: 10.1109/TKDE.2021.3109581.

[14] H. Yan, J. Li, B. Chu, and Z. Xu, "HT-STNet: a hierarchical Tucker decomposition and spatio-temporal LSTM network for accurate and efficient shared mobility demand forecasting on sparse data," *Appl. Intell.*, vol. 55, no. 7, p. 631, Apr. 2025, doi: 10.1007/s10489-025-06500-7.

[15] R. Madbhavi, B. Natarajan, and B. Srinivasan, "Enhanced Tensor Completion Based Approaches for State Estimation in Distribution Systems," *IEEE Trans. Ind. Inform.*, vol. 17, no. 9, pp. 5938–5947, Sep. 2021, doi: 10.1109/TII.2020.3035449.

[16] F. Cong, Q.-H. Lin, L.-D. Kuang, X.-F. Gong, P. Astikainen, and T. Ristaniemi, "Tensor decomposition of EEG signals: A brief review," *J. Neurosci. Methods*, vol. 248, pp. 59–69, Jun. 2015, doi: 10.1016/j.jneumeth.2015.03.018.

[17] H. Zhao, L. T. Yang, Z. Yang, D. Liu, X. Nie, and B. Ren, "Sparse Bayesian Tensor Completion for Data Recovery in Intelligent IoT Systems," *IEEE Internet Things J.*, vol. 11, no. 15, pp. 25682–25693, Aug. 2024, doi: 10.1109/JIOT.2024.3378202.

[18] X. Luo, S. Cheng, L. Wang, Y. Liang, and F. Lu, "A tensor decomposition method based on embedded geographic meta-knowledge for urban traffic flow imputation," *Int. J. Geogr. Inf. Sci.*, vol. 39, no. 4, pp. 788–816, Apr. 2025, doi: 10.1080/13658816.2024.2434665.

[19] X. Chen, Z. He, and L. Sun, "A Bayesian tensor decomposition approach for spatiotemporal traffic data imputation," *Transp. Res. Part C Emerg. Technol.*, vol. 98, pp. 73–84, Jan. 2019, doi: 10.1016/j.trc.2018.11.003.

[20] X. Chen, J. Yang, and L. Sun, "A nonconvex low-rank tensor completion model for spatiotemporal traffic data imputation," *Transp. Res. Part C*

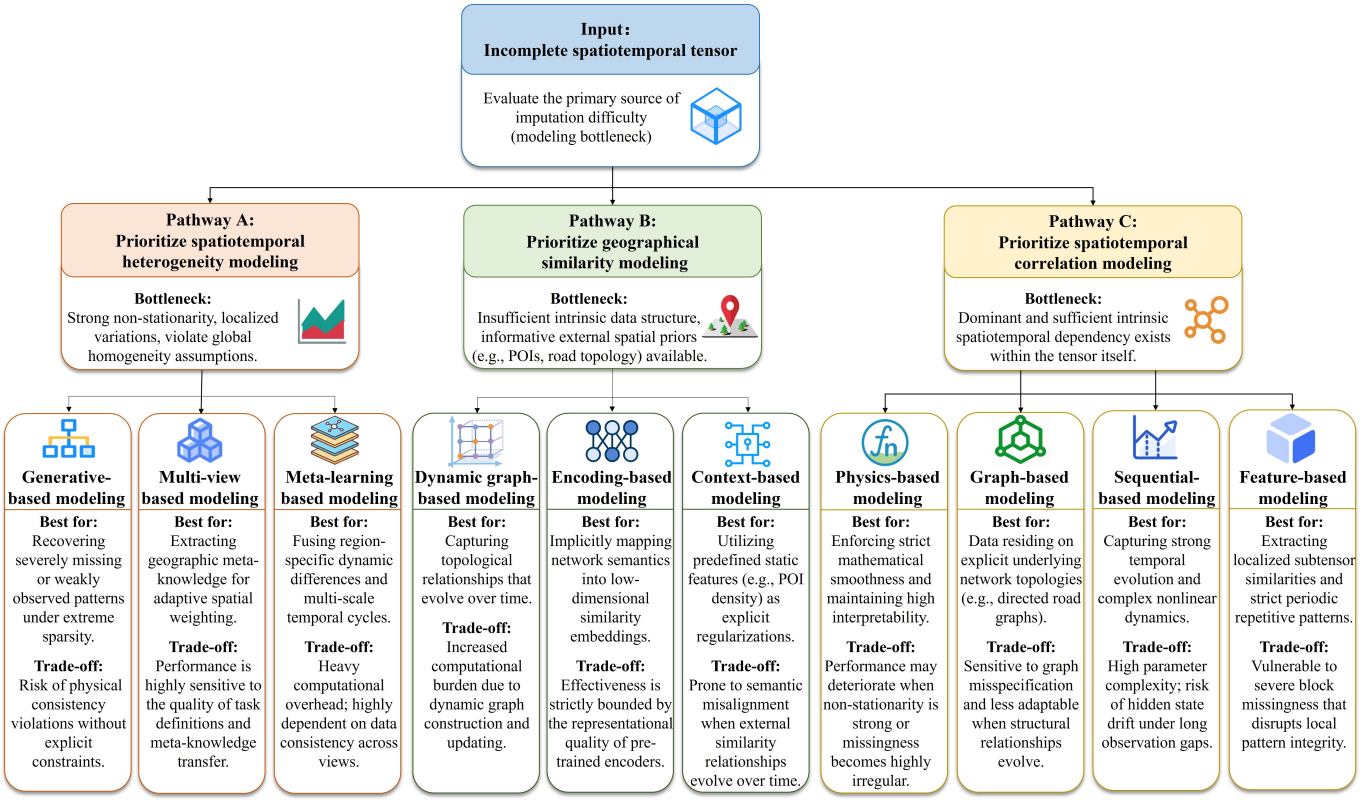

Fig. 1: Priority-based method decision flowchart.

Emerg. Technol., vol. 117, Aug. 2020, doi: 10.1016/j.trc.2020.102673.

[21] K. Zhang, L. Wu, L. Zheng, N. Xie, and Z. He, "Large-Scale Traffic Data Imputation with Spatiotemporal Semantic Understanding," Jan. 27, 2023, arXiv:2301.11691, doi: 10.48550/arXiv.2301.11691.

[22] R. Minster, I. Viviano, X. Liu, and G. Ballard, "CP decomposition for tensors via alternating least squares with QR decomposition," *Numer. Linear Algebra Appl.*, vol. 30, no. 6, p. e2511, 2023, doi: 10.1002/nla.2511.

[23] X. Luo, S. Cheng, Y. Yang, P. Wang, S. Guo, and F. Lu, "Heterogeneity-Guided Tensor Decomposition for Traffic Flow Imputation," *IEEE Trans. Intell. Transp. Syst.*, pp. 1–18, 2026, doi: 10.1109/TITS.2026.3658186.

[24] Y. He, Y. Jia, Y. Jia, C. An, Z. Lu, and J. Xia, "An Integrated Intra-View and Inter-View Framework for Multiple Traffic Variable Data Simultaneous Recovery," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 17200–17217, Nov. 2024, doi: 10.1109/TITS.2024.3414506.

[25] X. Chen, K. Wang, F. Zhao, F. Deng, and Q. Ye, "Composite Nonconvex Low-Rank Tensor Completion With Joint Structural Regression for Traffic Sensor Networks Data Recovery," *IEEE Trans. Comput. Soc. Syst.*, vol. 11, no. 5, pp. 6882–6896, Oct. 2024, doi: 10.1109/TCSS.2024.3406629.

[26] P. Chen, F. Li, D. Wei, and C. Lu, "Spatiotemporal traffic data completion with truncated minimax-concave penalty," *Transp. Res. Part C Emerg. Technol.*, vol. 164, Jul. 2024, doi: 10.1016/j.trc.2024.104657.

[27] Z. Zhao, L. Tang, M. Fang, X. Yang, C. Li, and Q. Li, "Toward urban traffic scenarios and more: a spatio-temporal analysis empowered low-rank tensor completion method for data imputation," *Int. J. Geogr. Inf. Sci.*, vol. 37, no. 9, pp. 1936–1969, Sep. 2023, doi: 10.1080/13658816.2023.2234434.

[28] X. Wang, Y. Wu, D. Zhuang, and L. Sun, "Low-Rank Hankel Tensor Completion for Traffic Speed Estimation," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 4862–4871, May 2023, doi: 10.1109/TITS.2023.3247961.

[29] T. Nie, G. Qin, Y. Wang, and J. Sun, "Correlating sparse sensing for large-scale traffic speed estimation: A Laplacian-enhanced low-rank tensor kriging approach," *Transp. Res. Part C Emerg. Technol.*, vol. 152, no. 15, p. 2, Jul. 2023, doi: 10.1016/j.trc.2023.104190.

[30] P. Chen, F. Li, D. Wei, and C. Lu, "Low-Rank and Deep Plug-and-Play Priors for Missing Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 2, pp. 2690–2706, Feb. 2025, doi: 10.1109/TITS.2024.3493864.

[31] H. Shu, J. Li, T. Lei, and L. Sun, "Robust Tensor Completion via Gradient Tensor Nuclear ℓ_1-ℓ_2 Norm for Traffic Data Recovery," *IEEE Trans. Intell. Transp. Syst.*, pp. 1–13, 2026, doi: 10.1109/TITS.2026.3659755.

[32] L. Yu, C. Guan, H. Wang, Y. He, W. Cao, and C.-S. Leung, "Robust Tensor Ring Decomposition for Urban Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 6, pp. 8707–8719, Jun. 2025, doi: 10.1109/TITS.2025.3555449.

[33] T. Nie, G. Qin, W. Ma, Y. Mei, and J. Sun, "ImputeFormer: Low Rankness-Induced Transformers for Generalizable Spatiotemporal Imputation," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, Barcelona, Spain: ACM, Aug. 2024, pp. 2260–2271, doi: 10.1145/3637528.3671751.

[34] H. Gao et al., "Identifying critical road sections using the weighted periodicity mined from trajectory data for efficient urban transportation," *GIScience Remote Sens.*, vol. 62, no. 1, p. 2448251, Dec. 2025, doi: 10.1080/15481603.2024.2448251.

[35] J. Zhang and J. Wei, "Multi-residual tensor completion for spatiotemporal data imputation," *Internet Things*, vol. 25, Apr. 2024, doi: 10.1016/j.iot.2024.101114.

[36] C. Yang, N. Singh, C. Xiao, C. Qian, E. Solomonik, and J. Sun, "MTC: Multiresolution Tensor Completion from Partial and Coarse Observations," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, Virtual Event, Singapore: ACM, Aug. 2021, pp. 1953–1963, doi: 10.1145/3447548.3467261.

[37] Z. Zhang, M. Li, X. Lin, and Y. Wang, "Network-wide traffic flow estimation with insufficient volume detection and crowdsourcing data," *Transp. Res. Part C Emerg. Technol.*, vol. 121, p. 102870, Dec. 2020, doi: 10.1016/j.trc.2020.102870.

[38] Q. Wang, L. Chen, Q. Wang, H. Zhu, and X. Wang, "Anomaly-Aware Network Traffic Estimation via Outlier-Robust Tensor Completion," *IEEE Trans. Netw. Serv. Manag.*, vol. 17, no. 4, pp. 2677–2689, Dec. 2020, doi: 10.1109/TNSM.2020.3024932.

[39] Y. Han and Z. He, "Simultaneous Incomplete Traffic Data Imputation and Similarity Pattern Discovery with Bayesian Nonparametric Tensor Decomposition," *J. Adv. Transp.*, vol. 2020, no. 10, pp. 1–25, Aug. 2020, doi: 10.1155/2020/8810753.

[40] Z. Zeng, B. Liu, J. Feng, and X. Yang, "Low-Rank Tensor and Hybrid Smoothness Regularization-Based Approach for Traffic Data Imputation With Multimodal Missing," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 10, pp. 13014–13026, Oct. 2024, doi: 10.1109/TITS.2024.3440011.

[41] H. Wu, L. Hu, X. Li, C. Wang, and Z. Ye, "A novel ship trajectory reconstruction approach based on low-rank tensor completion," *Ocean Eng.*, vol. 310, p. 118724, Oct. 2024, doi: 10.1016/j.oceaneng.2024.118724.

[42] X. Chen, K. Wang, F. Zhao, F. Deng, and Q. Ye, "Composite Nonconvex Low-Rank Tensor Completion With Joint Structural Regression for Traffic Sensor Networks Data Recovery," *IEEE Trans. Instrum. Meas.*, Jun. 2024, doi: 10.1109/TIM.2024.3406629.

[43] P.-L. Wu, M. Ding, and Y.-B. Zheng, "Spatiotemporal traffic data imputation by synergizing low tensor ring rank and nonlocal subspace regularization," *IET Intell. Transp. Syst.*, vol. 17, no. 9, pp. 1908–1923, Sep. 2023, doi: 10.1049/itr2.12383.

[44] M. Farazmand and A. K. Saibaba, "Tensor-based flow reconstruction from optimally located sensor measurements," *J. Fluid Mech.*, vol. 962, p. A27, May 2023, doi: 10.1017/jfm.2023.269.

[45] S. E. Sofuoglu and S. Aviyente, "GLOSS: Tensor-based anomaly detection in spatiotemporal urban traffic data," *Signal Process.*, vol. 192, p. 108370, Mar. 2022, doi: 10.1016/j.sigpro.2021.108370.

[46] A. Ghasemkhani, I. Niazazari, Y. Liu, H. Livani, V. A. Centeno, and L. Yang, "A Regularized Tensor Completion Approach for PMU Data Recovery," *IEEE Trans. Smart Grid*, vol. 12, no. 2, pp. 1519–1528, Mar. 2021, doi: 10.1109/TSG.2020.3030566.

[47] H. Chen, M. Lin, J. Liu, H. Yang, C. Zhang, and Z. Xu, "NT-DPTC: A non-negative temporal dimension preserved tensor completion model for missing traffic data imputation," *Inf. Sci.*, vol. 653, Jan. 2024, doi: 10.1016/j.ins.2023.119797.

[48] X. Xu, M. Lin, X. Luo, and Z. Xu, "HRST-LR: A Hessian Regularization Spatio-Temporal Low Rank Algorithm for Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 11001–11017, Oct. 2023, doi: 10.1109/TITS.2023.3279321.

[49] Z. Zeng, J. Feng, Z. Huang, B. Liu, and B. Zhou, "A Flexible Approach Based on Hybrid Global Low-Rankness and Smoothness Regularization With Nonlocal Structure for Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 12, no. 5, pp. 1–16, 2025, doi: 10.1109/TITS.2025.3541448.

[50] H. Chen, M. Lin, L. Zhao, Z. Xu, and X. Luo, "Fourth-Order Dimension Preserved Tensor Completion With Temporal Constraint for Missing Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 5, pp. 6734–6748, May 2025, doi: 10.1109/TITS.2025.3531221.

[51] Z. Zhao, L. Tang, C. Ren, X. Yang, Z. Kan, and Q. Li, "Diagnosing urban traffic anomalies by integrating geographic knowledge and tensor theory," *GIScience Remote Sens.*, vol. 61, no. 1, pp. 1–29, Dec. 2024, doi: 10.1080/15481603.2023.2290347.

[52] H. Shu, H. Wang, J. Peng, and D. Meng, "Low-Rank Tensor Completion With 3-D Spatiotemporal Transform for Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 11, pp. 18673–18687, Nov. 2024, doi: 10.1109/TITS.2024.3422214.

[53] L. Cheng, Q.-L. Lu, X. Wu, and C. Antoniou, "Tucker factorization-based tensor completion for robust traffic data imputation," *Transp. Res. Part C Emerg. Technol.*, vol. 160, Mar. 2024, doi: 10.1016/j.trc.2024.104502.

[54] X. Chen, Z. Cheng, H. Cai, N. Saunier, and L. Sun, "Laplacian Convolutional Representation for Traffic Time Series Imputation," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 11, pp. 6490–6502, Nov. 2024, doi: 10.1109/TKDE.2024.3419698.

[55] R. Mao, Z. Chen, and G. Hu, "Robust temporal low-rank representation for traffic data recovery via fused lasso," *IET Intell. Transp. Syst.*, vol. 15, no. 2, pp. 175–186, 2021, doi: 10.1049/itr2.12010.

[56] S. E. Sofuoglu and S. Aviyente, "GLOSS: A Tensor Decomposition Approach for Anomaly Detection in Spatiotemporal Data," Oct. 06, 2020, arXiv:2010.02889, doi: 10.48550/arXiv.2010.02889.

[57] H. Yan, J. Li, B. Chu, and Z. Xu, "HT-STNet: a hierarchical Tucker decomposition and spatio-temporal LSTM network for accurate and efficient shared mobility demand forecasting on sparse data," *Appl. Intell.*, vol. 55, no. 7, p. 631, Apr. 2025, doi: 10.1007/s10489-025-06500-7.

[58] D. Zhou, A. Uddin, Z. Shang, C. Sylla, X. Tao, and D. Yu, "A Fast Non-Linear Coupled Tensor Completion Algorithm for Financial Data Integration and Imputation," in *4th ACM International Conference on AI in Finance*, Brooklyn, NY, USA: ACM, Nov. 2023, pp. 409–417, doi: 10.1145/3604237.3626899.

[59] S. Cheng, P. Peng, and F. Lu, "A lightweight ensemble spatiotemporal interpolation model for geospatial data," *Int. J. Geogr. Inf. Sci.*, vol. 34, no. 9, pp. 1849–1872, Sep. 2020, doi: 10.1080/13658816.2020.1725016.

[60] K. Xie et al., "Neural Tensor Completion for Accurate Network Monitoring," in *IEEE INFOCOM 2020 - IEEE Conference on Computer Communications*, Jul. 2020, pp. 1688–1697, doi: 10.1109/INFOCOM41043.2020.9155366.

[61] J.-M. Yang, Z.-R. Peng, and L. Lin, "Real-time spatiotemporal prediction and imputation of traffic status based on LSTM and Graph Laplacian regularized matrix factorization," *Transp. Res. Part C Emerg. Technol.*, vol. 129, p. 103228, Aug. 2021, doi: 10.1016/j.trc.2021.103228.

[62] L. Deng, X.-Y. Liu, H. Zheng, X. Feng, and Y. Chen, "Graph Spectral Regularized Tensor Completion for Traffic Data Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 10996–11010, 2022, doi: 10.1109/TITS.2021.3098637.

[63] Van An Le et al., "GCRINT: Network Traffic Imputation Using Graph Convolutional Recurrent Neural Network," presented at the IEEE International Conference on Communications, 2021, doi: 10.1109/ICC42927.2021.9500687.

[64] Z. Zhang, H. Wang, Z. Fan, X. Song, and R. Shibasaki, "Missing Road Condition Imputation Using a Multi-View Heterogeneous Graph Network From GPS Trajectory," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 4917–4931, May 2023, doi: 10.1109/TITS.2023.3243087.

[65] X. Wu, M. Xu, J. Fang, and X. Wu, "A Multi-Attention Tensor Completion Network for Spatiotemporal Traffic Data Imputation," *IEEE Internet Things J.*, vol. 9, no. 20, pp. 20203–20213, Oct. 2022, doi: 10.1109/JIOT.2022.3171780.

[66] Y. Gong, Z. Li, J. Zhang, W. Liu, Y. Yin, and Y. Zheng, "Missing Value Imputation for Multi-View Urban Statistical Data via Spatial Correlation Learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 686–698, Jan. 2023, doi: 10.1109/TKDE.2021.3072642.

[67] L. Cai, C. Sha, J. He, and S. Yao, "Spatial-Temporal Data Imputation Model of Traffic Passenger Flow Based on Grid Division," *ISPRS Int. J. Geo-Inf.*, vol. 12, no. 1, Jan. 2023, doi: 10.3390/ijgi12010013.

[68] K. Xie et al., "Deep Adversarial Tensor Completion for Accurate Network Traffic Measurement," *IEEE/ACM Trans. Netw.*, vol. 31, no. 5, pp. 2101–2116, Oct. 2023, doi: 10.1109/TNET.2022.3233908.

[69] L. Han, K. Zheng, L. Zhao, X. Wang, and H. Wen, "Content-Aware Traffic Data Completion in ITS Based on Generative Adversarial Nets," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 11950–11962, Oct. 2020, doi: 10.1109/TVT.2020.3007025.

[70] L. Huang, Y. Yang, H. Chen, Y. Zhang, Z. Wang, and L. He, "Context-aware road travel time estimation by coupled tensor decomposition based on trajectory data," *Knowl.-Based Syst.*, vol. 245, p. 108596, Jun. 2022, doi: 10.1016/j.knosys.2022.108596.

[71] A. Ben Said and A. Erradi, "Spatiotemporal Tensor Completion for Improved Urban Traffic Imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6836–6849, Jul. 2022, doi: 10.1109/TITS.2021.3062999.

[72] J. Wang, J. Wu, Z. Wang, F. Gao, and Z. Xiong, "Understanding Urban Dynamics via Context-Aware Tensor Factorization with Neighboring Regularization," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 11, pp. 2269–2283, Nov. 2020, doi: 10.1109/TKDE.2019.2915231.

[73] R. Ouyang, Y. Hu, H. Wang, R. Hu, W. Yang, and K. Li, "A Novel Completion Method for Sparse Traffic Data Imputation," *IEEE Intell. Transp. Syst. Mag.*, pp. 2–14, 2025, doi: 10.1109/MITS.2024.3523353.

[74] A. Baggag et al., "Learning Spatiotemporal Latent Factors of Traffic via Regularized Tensor Factorization: Imputing Missing Values and Forecasting," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 6, pp. 2573–2587, 2021, doi: 10.1109/TKDE.2019.2954868.

[75] Z. Zhao, L. Tang, M. Fang, X. Yang, C. Li, and Q. Li, "Toward urban traffic scenarios and more: a spatio-temporal analysis empowered low-rank tensor completion method for data imputation," *Int. J. Geogr. Inf. Sci.*, vol. 37, no. 9, pp. 1936–1969, Sep. 2023, doi: 10.1080/13658816.2023.2234434.